

Modeling and Simulating Origami Structures using Bilinear Solid-Shell Element

Qixin Liang

The University of Hong Kong

Hong Kong, Hong Kong

TransGP

New Territories, Hong Kong

liangqixin23@outlook.com

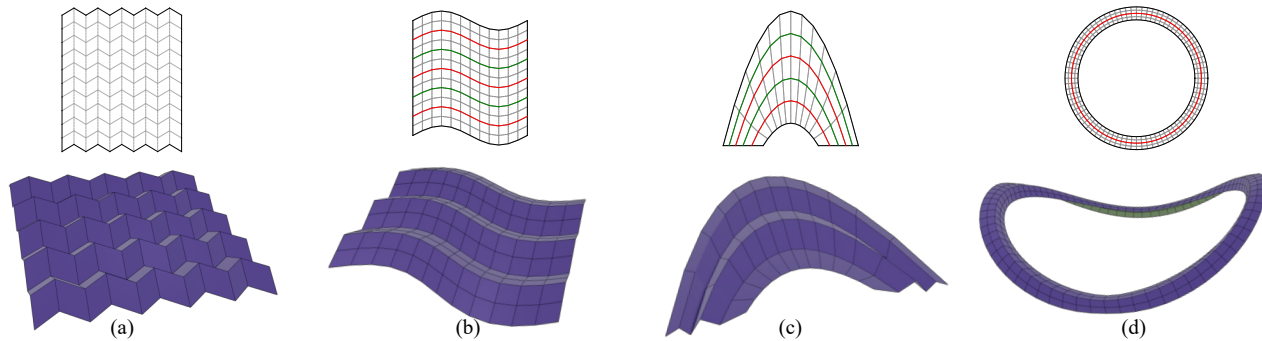


Figure 1: Our framework enables the modeling and simulation of origami structures with pre-creased straight and curved fold lines using bilinear solid-shell elements, as demonstrated in these images. This approach provides an effective framework for the design and analysis of origami structures.

Abstract

We propose a novel computational framework for modeling and simulating origami structures. In this framework, bilinear solid-shell elements are employed to model the origami panels while crease folding is considered through the angle between the director vectors of the adjacent panels. The director vector is the vector normal to the mid-surface before displacement/deformation comes in. To mitigate locking issues in the solid-shell element, we introduce the assumed natural strain method. To validate the effectiveness of our framework, we conduct origami simulations involving both straight- and curved-creases. The accuracy and efficacy of the framework are demonstrated through quantitative and qualitative analyses.

CCS Concepts

• **Computing methodologies** → **Physical simulation; Modeling and simulation.**

ACM Reference Format:

Qixin Liang. 2025. Modeling and Simulating Origami Structures using Bilinear Solid-Shell Element. In *SIGGRAPH Asia 2025 Technical Communications (SA Technical Communications '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3757376.3771398>

1 Introduction

Origami, rooted in the ancient art of paper folding, has evolved into a multidisciplinary field of science and engineering. As origami structures have transitioned from rigid panels to deformable panels exhibiting complex multi-physics responses, there is a growing need for simulation techniques that accurately capture both the geometric and physical behaviors.

Origami simulation methods can be broadly classified into kinematics-based and mechanics-based ones. *Kinematics-based* methods [Tachi 2009] assume panels remain rigid and planar, enabling folding angles only as variables to describe deformation. These methods are computationally efficient but cannot consider panel deformations. *Mechanics-based* methods relax the rigidity assumption, allowing membrane and bending deformations. The widely used bar-hinge model [Liu and Paulino 2017], similar to the mass-spring model in cloth simulation [Bridson et al. 2005], suffers from mesh-dependent issues, with material parameters that vary with mesh resolution and are not easily transferable across different mesh topologies. Discrete shell models [Burgoon et al. 2006] improve membrane accuracy of the origami panels but still exhibit mesh-dependent bending behavior. Other discrete and ruling-based models [Rabinovich et al. 2019; Solomon et al. 2012] impose strict isometry

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

SA Technical Communications '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-2136-6/25/12

<https://doi.org/10.1145/3757376.3771398>

constraints, making them difficult to consider physical constitutive laws. In [Hu et al. 2021], a corotational quadrilateral element was proposed to model the bending deformation of quadrilateral panels; however, it is limited to capturing only the warping bending within individual elements.

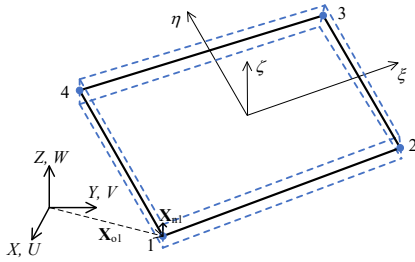
Motivated by these advancements, we introduce a solid-shell element to model origami panels. Instead of using discrete rods to represent deformed fold lines [Le et al. 2023], relying on remeshing to smooth sharp creases [Narain et al. 2013], or employing the normals of rigid prismatic panels [Kilian et al. 2017], we preserve the sharp features and conveniently model folding motion through the angle between director vectors of the elements along the same crease. Our contributions are:

- We propose a crease model based on the director vector of the solid-shell element.
- We present a total Lagrangian formulation for a bilinear quadrilateral solid-shell element [Sze et al. 2002], which effectively mitigates various locking phenomena through the assumed natural strain (ANS) method.
- We present two computational origami folding examples—one with a straight crease and the other with a curved crease—along with their analytical solutions, boundary conditions, and material setups. They may serve as useful references for benchmarking origami simulation tools.

This work aims to provide a versatile tool for the design and analysis of origami structures.

2 Computational model for origami simulation

2.1 Solid-Shell Finite Element Model



The above figure shows a bilinear solid-shell element with nodes on its mid-surface, the interpolated initial position vector $\mathbf{X}$ and displacement vector $\mathbf{U}$ can be written as:

$$\begin{aligned}\mathbf{X} &= \mathbf{X}_o(\xi, \eta) + \zeta \mathbf{X}_n(\xi, \eta) = \sum_i N_i(\xi, \eta) \mathbf{X}_{oi} + \zeta \sum_i N_i(\xi, \eta) \mathbf{X}_{ni}, \\ \mathbf{U} &= \mathbf{U}_o(\xi, \eta) + \zeta \mathbf{U}_n(\xi, \eta) = \sum_i N_i(\xi, \eta) \mathbf{U}_{oi} + \zeta \sum_i N_i(\xi, \eta) \mathbf{U}_{ni},\end{aligned}\quad (1)$$

where (ξ, η, ζ) is the natural coordinate vector ξ and its components are bounded by -1 and 1; N_i s are standard bilinear interpolation functions; $\mathbf{X}_o = [X_o, Y_o, Z_o]^T$ and $\mathbf{X}_n = [X_n, Y_n, Z_n]^T$ are the initial midsurface position and director vectors, while $\mathbf{U}_o = [U_o, V_o, W_o]^T$ and $\mathbf{U}_n = [U_n, V_n, W_n]^T$ are the midsurface and director displacement vectors.

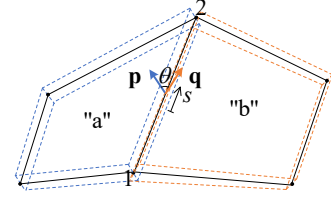
The initial nodal director $\mathbf{X}_{ni}$ should be taken to be perpendicular to the actual initial mid-surface which, unless is flat, is different from

the interpolated mid-surface $\mathbf{X}_o(\xi, \eta)$. The natural Green-Lagrangian strain tensor is

$$\boldsymbol{\varepsilon} = \frac{1}{2} \left(\left(\frac{\partial(\mathbf{X} + \mathbf{U})}{\partial \xi} \right)^T \frac{\partial(\mathbf{X} + \mathbf{U})}{\partial \xi} - \left(\frac{\partial \mathbf{X}}{\partial \xi} \right)^T \frac{\partial \mathbf{X}}{\partial \xi} \right). \quad (2)$$

Its components include $\varepsilon_{\phi\psi}$, $\gamma_{\zeta\phi}$ ($\phi, \psi = \xi, \eta$), and $\varepsilon_{\zeta\zeta}$, corresponding to the natural inplane, transverse shear, and transverse normal strains, respectively. More details about the element formulation and the assumed natural strain method are provided in the supplement material.

2.2 Crease Modeling



In the above Figure, $s \in [-1, 1]$ is the parametric coordinate of the crease between elements "a" and "b". While the crease is common, the directors of the two elements along the crease are independent. In the current configuration, let $\mathbf{p}$ and $\mathbf{q}$ denote the director of elements "a" and "b", respectively. They can be obtained by interpolating the deformed director $\mathbf{X}_{n1} + \mathbf{U}_{n1}$ at node 1, $s = -1$ and $\mathbf{X}_{n2} + \mathbf{U}_{n2}$ at node 2, $s = 1$, as

$$\begin{aligned}\mathbf{p} &= L_1(\mathbf{X}_{n1}^a + \mathbf{U}_{n1}^a) + L_2(\mathbf{X}_{n2}^a + \mathbf{U}_{n2}^a), \\ \mathbf{q} &= L_1(\mathbf{X}_{n1}^b + \mathbf{U}_{n1}^b) + L_2(\mathbf{X}_{n2}^b + \mathbf{U}_{n2}^b),\end{aligned}\quad (3)$$

where $L_1 = (1 - s)/2$ and $L_2 = (1 + s)/2$ are linear interpolation functions.

In thin shell analyses, transverse shear and thickness strain in the solid-shell strain elements are enforced to zero, the through-thickness direction is effectively inextensible, and the tangential relative motion of directors $\mathbf{p}$ and $\mathbf{q}$ along the crease is negligible. Therefore, the folding angle of crease can be approximated by

$$\theta = \begin{cases} \cos^{-1} \frac{\mathbf{p} \cdot \mathbf{q}}{|\mathbf{p}| |\mathbf{q}|} & \text{for } (\mathbf{p} \times \mathbf{q}) \cdot (\mathbf{x}_{o2} - \mathbf{x}_{o1}) \geq 0 \\ -\cos^{-1} \frac{\mathbf{p} \cdot \mathbf{q}}{|\mathbf{p}| |\mathbf{q}|} & \text{otherwise} \end{cases} \quad (4)$$

In the case that the two elements are flat, θ ranges from $[-\pi, \pi]$ with $\theta = \pm\pi$ indicating that the elements are fully folded. To prevent the physically inadmissible self-intersection configuration, the following crease energy Ψ_c with nonlinear folding constitutive model is adopted

$$\Psi_c = \begin{cases} \frac{1}{2} k_f (\theta_0 - \theta_L)^2 + k_f (\theta_0 - \theta_L) (\theta_L - \theta) - \frac{4k_f (\theta_L + \pi)^2}{\pi^2} \ln \left| \cos \left(\frac{\pi (\theta_L - \theta)}{2(\theta_L + \pi)} \right) \right| & -\pi < \theta < \theta_L \\ \frac{1}{2} k_f (\theta - \theta_0)^2 & \theta_L \leq \theta \leq \theta_R \\ \frac{1}{2} k_f (\theta_R - \theta_0)^2 + k_f (\theta_R - \theta_0) (\theta - \theta_R) - \frac{4k_f (\pi - \theta_R)^2}{\pi^2} \ln \left| \cos \left(\frac{\pi (\theta - \theta_R)}{2(\pi - \theta_R)} \right) \right| & \theta_R < \theta < \pi \end{cases}, \quad (5)$$

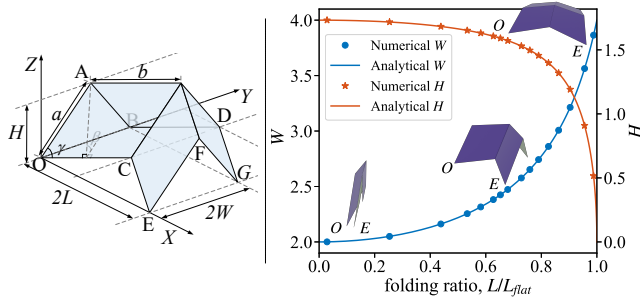


Figure 2: Compressing the Miura-ori unit cell. Left: A Miura-ori unit cell. Right: The width W and height H plotted against the folding ratio L/L_{flat} .

where $l = \|\mathbf{X}_{O2} - \mathbf{X}_{O1}\|$ is the crease length, and k_f is the folding stiffness per unit length [Liu and Paulino 2017]. The values θ_L and θ_R are fold angle limits that activate the penalty. Within the interval $[\theta_L, \theta_R]$, the energy exhibits a standard quadratic behavior centered at the rest angle θ_0 . Outside this range, the logarithmic term grows rapidly to impose strong penalty on configurations approaching $\theta = \pm\pi$, thereby effectively preventing self-intersection.

3 Results

Our computational model can be implemented in most, if not all, finite element programs. Here, we use a damped Newton solver with the adaptive increment method (see our supplemental document). In the tests below, SI units are used in expressing all material and geometric parameters. The quantitative validation includes two benchmark tests with known analytical solutions and an additional case presented in the supplemental document to further verify the solid-shell element. Subsequently, a series of qualitative tests is provided to illustrate the applicability of the present framework to more complex scenarios.

3.1 Quantitative tests

3.1.1 Compressing the Miura-ori unit cell. A Miura-ori unit cell is illustrated in the left image of Figure 2 and a complete Miura-ori structure is shown in Figure 1a. During the folding process, the cell deforms solely along the predefined creases. As a result, its deformation can be described analytically under the rigid panel assumption [Schenk and Guest 2013]. The analytical solution is given by

$$H = a \sin \beta \sin \gamma, \quad L = 2b \frac{\cos \gamma \tan \beta}{\sqrt{1 + \cos^2 \gamma \tan^2 \beta}}, \quad W = 2a \sqrt{1 - \sin^2 \gamma \sin^2 \beta}. \quad (6)$$

The geometry of the Miura-ori unit cell is defined by the parameters $a = 2$, $b = 2$, and $\gamma = 60^\circ$. To model the Miura-ori unit cell as a "rigid origami" system—consisting of rigid panels connected by compliant hinges—we follow the approach in [Liu and Paulino 2017], setting the bending rigidity of the panels to be 10^5 times greater than the folding stiffness. The material parameters used are $E = 12 \times 10^9$, $\nu = 0.3$, $h = 0.01$, and $k_f = 0.01$. To avoid buckling, β is set to be 15° in the initial configuration, making the Miura-ori unit cell nearly

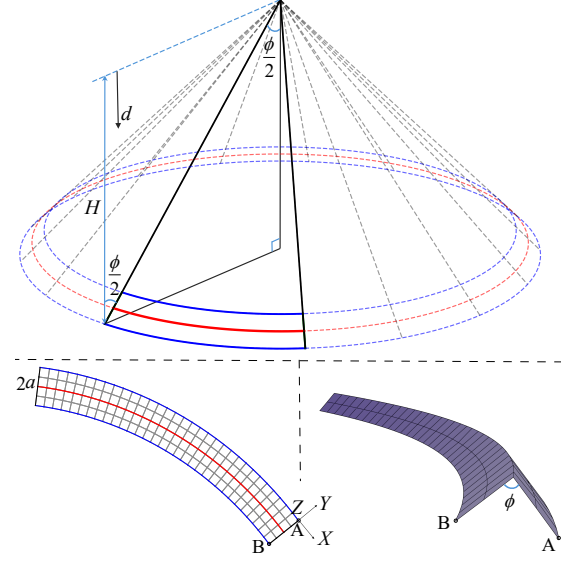


Figure 3: Folding a creased annulus sector into a theoretical cone. The 32×4 meshed cut annulus shown at the bottom left is extracted from the cone at the top, which has an apex angle of $\phi = 90^\circ$. The curved-crease origami structure on the bottom right also exhibits a fold angle of $\phi = 90^\circ$.

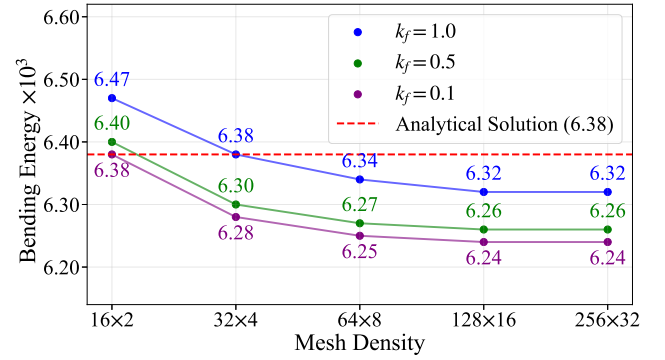


Figure 4: The bending energy in the cut annulus versus the mesh density.

flat. The boundary conditions are imposed as follows: $\mathbf{U}_O = \mathbf{0}$ at node O; the $U_o = 0$ at nodes A and B; the $W_o = 0$ at nodes O, B, C, D, E, and G; $U_o = -3.44$ is prescribed to nodes E, F, and G to compress the unit cell into a folded configuration. As shown in the right image of Figure 2, our predictions are indistinguishable from the analytical solution.

3.1.2 Folding a creased annulus sector into a theoretical cone. Following [Woodruff and Filipov 2020], we cut an annulus sector from a cone, flatten it into a planar sheet, and then fold it along the curved crease into a curved-crease structure (see Figure 3). Folding the initially flat sheet along a curved crease involves bending the developable cone surfaces during the motion. Under the isometric assumption, the theoretical bending energy of the annular region

on the right cone surface can be computed as

$$\Psi_{\text{annulus}} = \frac{1}{2} \int_{\Omega} D_b \kappa_p^2 d\Omega, \quad (7)$$

where $D_b = Eh^3/12(1 - \nu^2)$ is the bending rigidity of an isotropic shell, Ω is the area of the annular region, and the principal curvature κ_p is

$$\kappa_p = \frac{2}{\tan(\phi/2) \sqrt{1 + \tan^2(\phi/2)d}}, \quad (8)$$

where d is the distance from the apex of the cone measured along its height, and $\phi/2$ is half of the cone's apex angle (see Figure 3). Following [Woodruff and Filipov 2020], when the fold angle of the curved-crease structure is $\phi = 90^\circ$, the resulting folded shape becomes an exact segment of a cone.

In the numerical simulation, we adopt the geometric and material parameters from [Woodruff and Filipov 2020]. The radius of the middle arc is $R = 0.1$, which bisects the annulus of width $a = 0.005$, and the central angle is $\alpha = \pi/4$. $E = 4 \times 10^9$, $\nu = 0$, and $h = 0.1 \times 10^{-3}$. The folding stiffness k_f is typically scaled relative to the bending rigidity of the shell [Lechenault et al. 2014]; hence, we consider three representative values: 0.1, 0.5, and 1. The boundary conditions are imposed as follows: $\mathbf{U}_o = \mathbf{0}$ at node A; $V_o = W_o = 0$ at node B; $W_o = 0$ is also prescribed for all nodes along the inner and outer arcs; $W_o = a/\sqrt{2}$ is applied to the middle arc to induce a folding angle of 90° .

As shown in Figure 4, the predicted bending energy rapidly converges close to the theoretical value as the mesh is refined, with relative errors of approximately 0.9%, 1.8%, and 2.1% corresponding to folding stiffness values of 1, 0.5, and 0.1, respectively. This clearly illustrates the influence of the folding stiffness on the bending response. The case highlights the potential of our method for simulating and analyzing the curved-crease origami structures.

3.2 Qualitative tests

Figures 1 and 5 present additional qualitative results, illustrating our model's capability to handle more complex structures.

4 Conclusion and Discussion

In this paper, we propose the use of solid-shell elements to model origami structures, where the naturally embedded director vectors of the elements are utilized to compute the crease folding angle. Origami structures with straight- and curved-creases are considered. Our predictions are close to the analytical solutions.

Our modeling approach can be naturally extended to triangular and higher-order solid-shell elements, while still utilizing director vectors to model creases, thereby enabling the simulation of curved-tile origami structures [Liu and James 2024], which are challenging to capture with existing methods.

Acknowledgments

The author thanks Professor K.Y. Sze and the anonymous reviewers for their valuable comments, and gratefully acknowledges the funding supporter, TransGP.

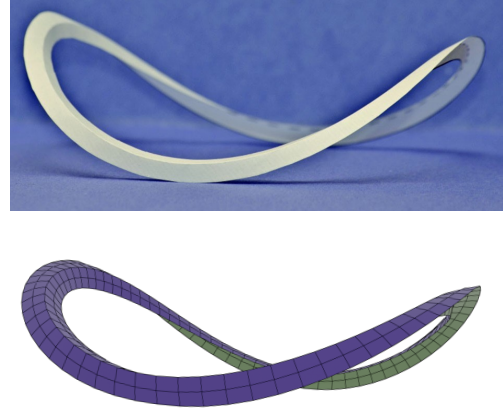


Figure 5: Buckle the paper-made full annulus. The top image is reprinted with permission from Marcelo A. Dias, Levi H. Dudte, L. Mahadevan, and Christian D. Santangelo, *Phys. Rev. Lett.* **109**, 114301 (2012). Copyright ©2012 American Physical Society. (DOI: <http://dx.doi.org/10.1103/PhysRevLett.109.114301>) The below one is our simulated result.

References

- R. Bridson, S. Marino, and R. Fedkiw. 2005. Simulation of clothing with folds and wrinkles. In *ACM SIGGRAPH 2005 Courses* (Los Angeles, California) (SIGGRAPH '05). Association for Computing Machinery, New York, NY, USA, 3–es. doi:10.1145/1198555.1198573
- Rob Burgoon, Zoë J. Wood, and Eitan Grinspun. 2006. Discrete Shells Origami. In *Proceedings of the 21st International Conference on Computers and Their Applications (CATA 2006)*. ISCA, Seattle, Washington, USA, 180–187. https://digitalcommons.calpoly.edu/csse_fac/204/
- YC Hu, YX Zhou, KW Kwok, and KY Sze. 2021. Simulating flexible origami structures by finite element method. *International Journal of Mechanics and Materials in Design* **17** (2021), 801–829. doi:10.1007/s10999-021-09538-w
- Martin Kilian, Aron Monszpart, and Niloy J. Mitra. 2017. String Actuated Curved Folded Surfaces. *ACM Trans. Graph.* **36**, 4 (July 2017), 13 pages. doi:10.1145/3072959.3015460
- Qiqin Le, Yitong Deng, Jiamu Bu, Bo Zhu, and Tao Du. 2023. Second-Order Finite Elements for Deformable Surfaces. In *SIGGRAPH Asia 2023 Conference Papers (SA '23)*. Association for Computing Machinery, New York, NY, USA, Article 113, 10 pages. doi:10.1145/3610548.3618186
- F. Lechenault, B. Thiria, and M. Adda-Bedia. 2014. Mechanical Response of a Creased Sheet. *Phys. Rev. Lett.* **112** (Jun 2014), 244301. Issue 24. doi:10.1103/PhysRevLett.112.244301
- Huan Liu and Richard D. James. 2024. Design of origami structures with curved tiles between the creases. *Journal of the Mechanics and Physics of Solids* **185** (2024), 105559. doi:10.1016/j.jmps.2024.105559
- Ke Liu and Glaucio H Paulino. 2017. Nonlinear mechanics of non-rigid origami: an efficient computational approach. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* **473**, 2206 (2017), 20170348. doi:10.1098/rspa.2017.0348
- Rahul Narain, Tobias Pfaff, and James F. O'Brien. 2013. Folding and crumpling adaptive sheets. *ACM Trans. Graph.* **32**, 4 (July 2013), 8 pages. doi:10.1145/2461912.2462010
- Michael Rabinovich, Tim Hoffmann, and Olga Sorkine-Hornung. 2019. Modeling curved folding with freeform deformations. *ACM Trans. Graph.* **38**, 6 (Nov. 2019), 12 pages. doi:10.1145/3355089.3356531
- Mark Schenk and Simon D. Guest. 2013. Geometry of Miura-folded metamaterials. *Proceedings of the National Academy of Sciences* **110**, 9 (2013), 3276–3281. doi:10.1073/pnas.1217998110
- Justin Solomon, Etienne Vouga, Max Wardetzky, and Eitan Grinspun. 2012. Flexible Developable Surfaces. *Computer Graphics Forum* **31**, 5 (2012), 1567–1576. doi:10.1111/j.1467-8659.2012.03162.x
- K. Y. Sze, W. K. Chan, and T. H. H. Pian. 2002. An eight-node hybrid-stress solid-shell element for geometric non-linear analysis of elastic shells. *Internat. J. Numer. Methods Engrg.* **55**, 7 (2002), 853–878. doi:10.1002/nme.535
- Tomohiro Tachi. 2009. Simulation of rigid origami. *Origami* **4**, 08 (2009), 175–187.
- Steven R. Woodruff and Evgueni T. Filipov. 2020. A bar and hinge model formulation for structural analysis of curved-crease origami. *International Journal of Solids and Structures* **204–205** (2020), 114–127. doi:10.1016/j.ijsolstr.2020.08.010